

Java OS + JVM — Visual Diagram Sheet

Q1 — Process Flow

Terminal	OS Process	JVM Runtime	Your Java App
bash/zsh calls OS	Allocates memory Creates process	Loads classes Creates system threads	main() executes

Q2 — Threads Before main()

OS Layer	JVM Internal Threads	Application Thread
Bootstrap Thread	GC Threads JIT Threads VM Thread Signal Dispatcher Reference Handler	main() (starts after init)

Q3 — Production Thread Classification

Application Threads	Framework Threads	JVM Internal Threads
ExecutorService @Async Custom pools Your package in stack trace	Tomcat http-nio-* Netty nioEventLoopGroup-* ForkJoinPool Scheduler-*	GC Thread VM Thread CompilerThread Signal Dispatcher